

Xu-Cheng Wang

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EDUCATION

Fudan University, Department of Physics

PhD in Theoretical Physics (Condensed Matter Theory)

– Elite PhD scholarship, GPA: 3.83/4.0

– Supervised by Prof. Yang Qi

– Thesis title: *Preformed pairs and phase fluctuations in two-dimensional superconductors*

Shanghai, China

Sept. 2021 – Present

Fudan University, Department of Physics

B.S. in Physics, GPA: 3.65/4.0 (top 15%)

Shanghai, China

Sept. 2017 – Jun. 2021

ACADEMIC RESEARCH

Phase fluctuations, charge and magnetic pseudogap in 2D superconductors

- We theoretically studied the phase fluctuations in generic two-dimensional superconductors and predicted the emergence of charge pseudogap, Fermi arcs, and magnetic pseudogap without invoking competing order.
- We validated these predictions by simulating strongly correlated models using determinant quantum Monte Carlo. The associated DQMC and SAC implementations are available as open-source code at [dqmc-framework](#) and [sac](#).
- **Related papers:**
 1. **X.-C. Wang** and Yang Qi, *Phase fluctuations in two-dimensional superconductors and pseudogap phenomenon*, [Phys. Rev. B](#), **107**, 224502 (2023).
 2. **X.-C. Wang**, Xiao Yan Xu, and Yang Qi, *The interplay of phase fluctuations and nodal quasiparticles: ubiquitous Fermi arcs in two-dimensional d-wave superconductors*, [Sci. Bull.](#) **71**, 36-39 (2026).
 3. **X.-C. Wang** and Yang Qi, *Emergent magnetic pseudogap from phase fluctuations and hierarchy of scales in two-dimensional superconductors*, [arXiv:2608.07238](#) (2026).

Quantum geometric localization length in an ideally flat Chern band

- We showed that the localization length in an isolated, ideally flat Chern band is a quantum geometric length. The quantum geometry also governs the criticality of localization transition and drives a flow of critical exponents connecting the Dirac and orthogonal fixed points.
- Our findings provide a novel quantum geometric perspective on the localization in quantum Hall systems and shed new light on the long-standing controversy over the criticality of the integer quantum Hall transition.
- **X.-C. Wang** and Yang Qi, *Quantum geometric localization length and localization criticality in an ideally flat Chern band*, [arXiv:2608.08042](#) (2026).

Nullspace-guided adaptive bootstrap of quantum many-body systems

- The many-body bootstrap is a relaxation approach that yields rigorous lower bounds on ground-state energies by optimizing over an outer approximation to the physical feasible set. We developed the nullspace-guided adaptive bootstrap, which dynamically refines the operator basis using the moment matrix nullspace and iteratively improves the energy bound.
- The method yields nearly exact ground-state energy bounds for the transverse-field Ising chain and improves state-of-the-art lower bounds for the Hubbard chain by up to two orders of magnitude. It provides a practical and scalable route towards accurate certified bounds on the ground-state energies and general observables of quantum many-body systems. An open-source implementation is available at [QMBBoot-NGA](#).
- **X.-C. Wang** and Yang Qi, *Nullspace-guided adaptive bootstrap of quantum many-body systems*, [to appear](#) (2026).

LANGUAGES & TECHNICAL SKILLS

Languages: Mandarin, English

Programming: C/C++, Python, Linux, HPC, AI Agents